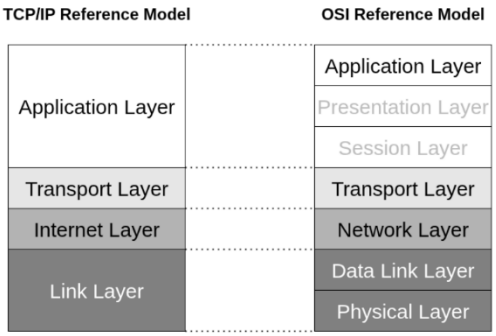
Propagation Delay: Bis das ganze Auto raus ist > first bit => size / bandwith
Transmission delay: Wie lange die ganze reise dauert für 1 bit => distance / speed
Gesamt: Prop. + Trans.
Processing delay: processing of router / host / switch (const var)
Queuing delay: How long stuff waits in queue before sent
RTT = t_to_reciever + t_to_sender (t * 2 most of the time)

Layer	Name	Example Technology / Protocol	PDU name	One Function of the Layer
7	Application	HTTP, FTP, SMTP, DNS	**Data**	Provides network services directly to user applications
6	Presentation	TLS/SSL, JPEG, MPEG	**Data**	Data representation (encryption, compression, encoding)
5	Session	NetBIOS Session Service, RPC	**Data**	Session establishment, management, and termination
4	Transport	TCP, UDP	**Segment** (TCP) / **Datagram** (UDP)	End-to-end communication, flow & error control
3	Network	IP, ICMP, IPsec	**Packet**	Logical addressing and routing between networks
2	Data Link	Ethernet (IEEE 802.3), Wi-Fi, ARP	**Frame**	Framing, MAC addressing, error detection
1	Physical	Ethernet PHY, RS-232, fiber optics	**Bit**	Transmission of raw bits over the physical medium



Bereich	Erkennung
Multicast	224–239.x.x.x
Broadcast	255.255.255.255
Loopback	127.x.x.x
Link-Local	169.254.x.x
Private	10.x.x.x / 172.16–31.x.x / 192.168.x.x

Bereich	Präfix	Bedeutung	Geroutet?
Global Unicast	2000::/3	Öffentliche IPv6-Adresse	Ja
Unique Local (ULA)	fc00::/7 (meist fd..)	Privat	Nein
Link-Local	fe80::/10	Nur lokales Netz	Nein
Multicast	ff00::/8	Multicast	Nein
Loopback	::1	Loopback	Nein

Msg	ACK	SYN	FIN	Payload len	Seq	Ack
1	0	1	0	0	3247	0
2	1	1	0	0	7	3248
3	1	0	0	0	3248	8
Msg	**ACK**	**SYN**	**FIN**	**Payload len**	**Seq**	**Ack**
4	0	0	0	130	50	500
5	1	0	0	200	500	180
6	1	0	0	130	180	700
7	1	0	0	20	700	310
Msg	**ACK**	**SYN**	**FIN**	**Payload len**	**Seq**	**Ack**
8	0	0	1	0	1	2342
9	1	0	0	0	2342	2
10	0	0	1	0	2342	2
11	1	0	0	0	2	2343

Baud: signals per second
NRZ: |-----| <- geht nicht wieder runter bei mehreren gleich; Line Code: _ = 0, - = 1 (das mapping selber, NRZ ist ein line code)

Message 1: 10.0.0.4:444555	193.104.220.7:433
Message 2: 194.95.83.45:56478	193.104.220.7:443
Message 3: 193.104.220.7:443	194.95.83.45:56478
Message 4: 193.104.220.7:443	10.0.0.4:444555

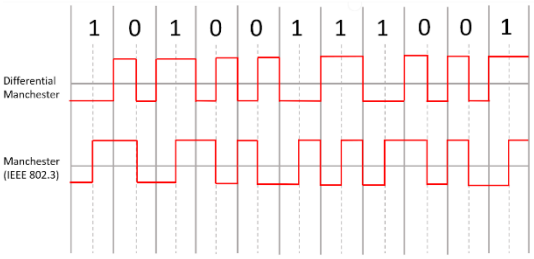
8-Bit payload => 7 Bits are beeing appended. (app = len(payload) – 1)
Message: 11010011 0000000; Errors detect: d – 1; Errors correct: floor((d-1) / 2)

```
110100110000000 ^
100101
 100011
 100101
   110100
   100101
    100010
    100101
      111000
      100101
       111010
       100101
       ------
       011111
```

POTS frequence: 3100Hz; SNR = P_signal / P_noise
C = B * log2(1 + SNR); C: bit/s, B: Hz, SNR: -

Line Coding ist für **Basisband-Übertragung** gedacht, aber die Telefonleitung ist ein **Bandpasskanal** (ca. 300–3400 Hz), daher müssen die Daten per **Modulation auf eine Trägerfrequenz** gelegt werden, um übertragen werden zu können.

Bit Stuffing: When HDLC discovers 5x 1 bits in the payload / body, it puts a 0 at the end of the 5x 1s. This is undone upone recieving.



NRZ has the problem that no signal change can bring the listener / reciever out of tact / rythm when transmitting the same type of bits back to back (**Baseline Wanderer**). Manchester solves this. The process of solving this problem without another tick channel is called **clock recovery.** The process of mapping electrical signals to bits is called **Modulation.** Indicating the bits not by voltage but by **change of voltage** (like manchester) is called **Alternative Mark Inversion (AMI).**

The **ARP - Adress Resolution Protocol** works by sending a broadcast containing the IPv4 Address of the target device. Each device compares this to its own IPv4 Address; If it matches, the device sends an ARP response via unicast containing its MAC address back to the sender. ARP only has one job: Mapping a MAC address to a IPv4 Address. **NDP** does alot more. ARP uses Broadcast instead of multicast like NDP, causing extreme security riscs and more network load. NDP uses SLAAC to determine an IPv6 address for you. NDP can find routers on its own. ARP is used for IPv4 and more, NDP is specifically designed for IPv6. (ARP was also designed for IPv4, but can be used for a lot of other things, unlike NDP)
Ethernet cable types
twisted pair cables (e.g. Cat5/Cat6), fiber-optic cables, and historically coaxial cables.
Difference between coaxial and twisted pair cables
Coaxial cables use a single shielded conductor, while twisted pair cables use two twisted wires to reduce electromagnetic interference.
Advantages of fiber-optic cables over copper
Fiber-optic cables provide higher bandwidth, longer transmission distances, and are immune to electromagnetic interference.
Transmission range vs. interference range
transmission range is the area where a signal can be correctly received; Interference range is where the signal can still disturb other comms.
Wireless network challenges
Wireless networks suffer from interference caused by other transmitters and signal attenuation due to distance and obstacles.
Why omnidirectional transmission is still used
Omnidirectional transmission is cheaper, simpler, and allows mobile devices to connect without precise antenna alignment.
Why attenuation affects wireless networks more
Wireless signals spread freely through space and are weakened by obstacles, while wired signals are guided and shielded inside cables.
Popular network technologies in companies or universities
Ethernet (LAN) and Wireless LAN (WLAN).
Communication model used by eduroam
Eduroam uses a client–server communication model with centralized authentication.
Frequency spectrum used by WLAN
WLAN operates mainly in the 2.4 GHz and 5 GHz ISM bands, and also in the 6 GHz band with Wi-Fi 6E.
Higher transmission power: Bluetooth or WLAN
WLAN achieves higher trans. power because it´s designed for high data rates and long range; Bluetooth is optimized for low energy.
IEEE 802.11b-WLAN has a data rate of 11Mbit/s

Method	*Used in*	*How it works*	*Needs server?*
Address Autoconfiguration	IPv4/IPv6	Auto IP, general term	maybe
DHCP	IPv4/IPv6	Ask server → get IP + info	Yes
SLAAC	IPv6	Listen Router Advertisement → make own IP	No

Protocol	*Type*	*How it works*	*Used in*
ALOHA	Random	Send anytime, if collision → try again	Early wireless, satellite
CSMA/CA	Random	Check channel, wait if busy, use RTS/CTS	Wi-Fi
MACA	Random	Use RTS/CTS to avoid hidden node collision	Wireless LAN
TDMA	Time	Each user has own time slot	GSM, cellular, satellite
FDMA	Frequency	Each user has own frequency band	Radio, old mobile
CDMA	Code	All use same frequency, each has unique code	3G mobile, GPS

Bridges simply connect 2 or a few more hosts directly together; **Switches** connect multiple, are faster and full duplex (bi-directional). **Forwarding tables** save the last send messages from host to host. If **no entry** for a host exists, the bridge floods to all connected ports except the reciever. **Learning bridges** forward data dynamically by observing source MAC addresses. Bridges use the **Spanning Tree Protocol STP** to avoid **loops. Root Bridge** has the lowest **Bridge ID** and highest **priority**.

Media Access Controle (MAC) is used to coordinate multiple devices without collision. The MAC number is burned into the hardware (48 bit global std; first 24bit manufac, other Seriesnum). Both ethernet and wlan are **non-determ.**, ether uses CSMA/CD, wlan uses CSMA/CA. /CD tries to detect collisions, CA tries to avoid them (like TCP slowstart). In CA **additional RTS / CTS frames** are used to reserve the medium and avoid collisions. A **higher p value** is used for low loads and higher more aggressive transmission with less delay.

An autonomous system (AS): collection of IP networks under a single administrative control that share a common routing **policy**.
Distance-vector: Bellman-Ford, knows neighbours and distance to every target.
Link-state: Dijkstra, knows the whole network. **(Both are Routing Protocol classes)**
BGP (Distance vec) is a protocol for Inter-AS routing, pathing is based on policy and neighbour AS‘s.
OSPF (link-state) is a protocol for Intra-AS routing, knows the whole network.
RIP (Distance vec) is a protocol for Intra-AS routing, only knows neighbour, paths based on **hops** to target.
In practice, both BGP and OSPF are used together.

Subnet ID = Network IP => BITWISE_AND(Mask, IP)
Package **stays in network** if (NetworkIP_reciever == NetworkIP_sender)
MTU: Maximum Transmission Unit; max size of package, if the size-to-be-send exceeds this, its **(IP-) fragmented**, meaning that every package gets its own header. => more overhead, if one fragment is lost, the whole package needs to be sent again.
ICMP: Internet Control Message Protocol; used when errors occur in IP-Networks (e.g. **TTL (Time to live)** ran out, package not reachable). Pings a **ICMP Echo Request** to the target => the target awnsers with **ICMP Echo Reply**; This tests if the host is reachable and how long the response takes.
Traceroute: uses incrementing TTL to discover route to target, resets when TTL reaches 0, starting the loop again.

Type	*Range / Purpose*
PAN	Personal Area Network, e.g., Bluetooth, few meters
BAN	Body Area Network, e.g., medical sensors on the body
LAN	Local Area Network, e.g., office or building, high speed
MAN	Metropolitan Area Network, e.g., city-wide network
WAN	Wide Area Network, e.g., Internet, connects cities/countries

Algorithm	*Order of the nodes being read*
Bellman-Ford	**No order,** iterates over all nodes till there are no changes in a full iteration.
Dijkstra	Node with the smalles distance to start-point is viewed next, uses priority queue.

Type	*Description*
Unicast	1-to-1 → sent to a single recipient
Broadcast	1-to-all → sent to all devices in the network
Multicast	1-to-group → sent only to a specific group of recipients
Anycast	1-to-nearest → sent to the nearest/optimal node in a group

Synchronous: sender and receiver are **time-synchronized**, data sent at fixed intervals. **Asynchronous** is not fixed.

Mode	*Description*
Simplex	one-way only → sender → receiver
Half-Duplex	both ways, but **alternating**
Full-Duplex	both ways **simultaneously**

TCP (with AIMD), UDP & QUIC:
Silly Window Syndrome occurs when small segments of data are sent inefficiently, causing excessive overhead and reduced network performance.
Avoidance techniques prevent sending very small segments until enough data is accumulated or the receiver's window is sufficiently large.
Two Windows are contained when using TCP, one for the network capacity and one for the capacity of the reciever.
Fast Recovery is optional and activated by the reciever by sending 3 ACKs back-to-back, causing the threshhold of the slow start to halve.
Thresholds start at big sizes, e.g. 64kB. Upon **timeout**, the cwnd (current window size) is dropped back to **1** (not zero!). When the threshhold is reached **without timeout**, TCP switches to the **congestion-avoidance-phase / linear mode**, setting MSS = 1kB. TCP does not stop „exploring“ the bounderies of the reciever.
TCP is **Connection-Oriented** (First Connection, then send data).
Congestion can occur upon exceeding the network capacity or by overloading routers / switches.
t_congestion = RTT * rounded_up(log2(size_window / size_segment))
UDP just sends stuff and forgets. (**Connectionless**)
Protocols which use UDP, like **DNS**, **switch back to TCP** when UDP isnt enough anymore (e.g. segment too big).
UDP is **used** when speed mathers more then security e.g. livestreams & videogames.
QUIC is a more secure protocol based on UDP. Developed by Google. Uses multiplexing and implements Head-of-Line-blocking and HTTPS (like TCP).
DNS maps **Domain** to **IP**, which is stored in the **Resource Record (RR)**. Each part of a domain („www“, „google“, „.com“, etc.) is called a **Label.**
The **Zone** is a subset of the DNS-Namespace, which controls other RRs of this domain. This is the **index** (google.com / youtube.com) most of the time.
FQDN: Full qualified Domain name, e.g. www.google.com
SYN Flood: Send giant amount of Syn requests to a host, causing it to open a lot of windows, blocking them for other users **(Denial of Service / DOS)**
The **hidden terminal problem** is a wireless networking collision issue where two nodes (A and C) outside each other's range both transmit to a commonrec incorrectly sense the channel as free.